HCPS: Heuristic Candidate-space Policy Search for Agile Earth Observation Satellite Scheduling

Pengfei Gao[1], Benkui Zhang[2], Yajing Xu[1], Jinxiao Wang[2], Zhiyuan Yu[2], Kangning Du[3], Bo Xiao[1]

[1] Beijing University of Posts and Telecommunications; [2] Key Lab of Target Cognition and Application Technology, Aerospace Information Research Institute; [3]Key Laboratory of Information and Communication Systems, Ministry of Information Industry, Beijing Information Science and Technology University

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Introduction & Problem

Advances in satellite technology have boosted the capabilities of Agile Earth Observation Satellites (AEOS), increasing the need for efficient mission scheduling. However, the growing number of satellites and their longer Visible Time Windows (VTWs) add significant complexity. Unlike traditional satellites, AEOS can adjust their roll, pitch, and yaw to extend observation times. While longer VTWs provide more options for actual Observation Windows (OWs), they also create more conflicts between windows, making the Agile Earth Observation Satellite Scheduling Problem (AEOSSP) a critical and challenging optimization task.

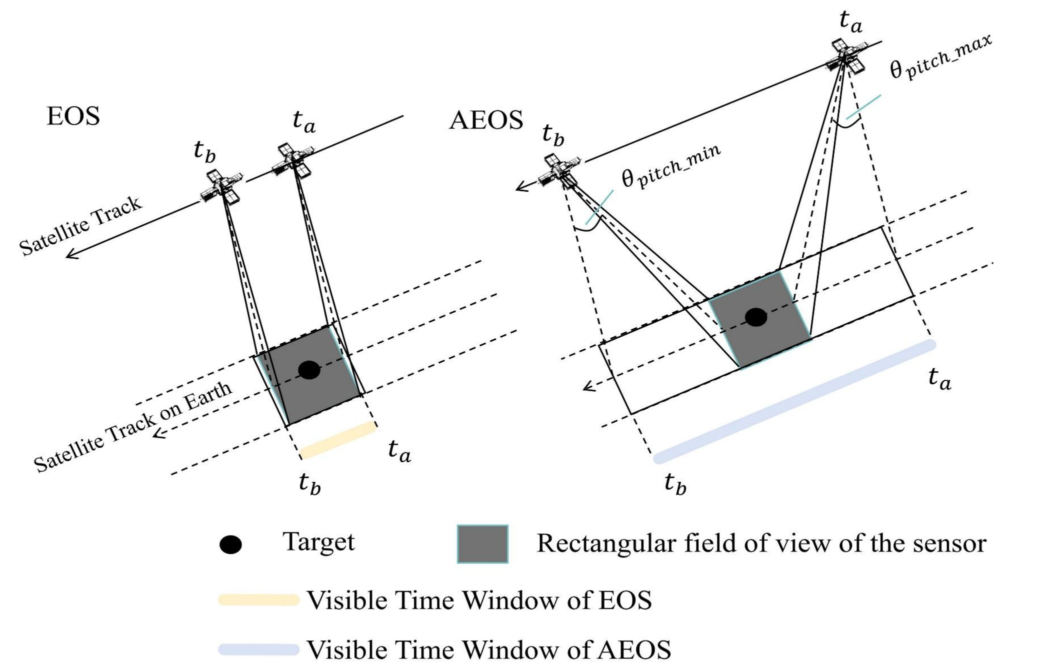

Figure 1: Comparison of VTWs for EOS and AEOS

Formulation of the AEOSSP

(1). Problem Constraints & Optimization Objective

- Each OW must lie within the VTW and meet the minimum duration.
- Attitude Transition: Sufficient time must be allocated between consecutive OWs for attitude maneuvering.
- Uniqueness: Each target can be observed at most once across all orbits.
- Maximize the total profit of all successfully scheduled observations.

(2). Decision variables

- Binary variable x_i^j : equals 1 if target T_i is observed in orbit O_j, otherwise 0.
- Real variable os_i^j: start time of the observation window OW_i^j when $x_i^j = 1$.

(3). Mathematical model

$$\begin{aligned}
\text{Maximize}\quad & \sum_{i\in T}\sum_{j\in O} p_i \cdot x_i^j \\
\text{subject to}\quad & \sum_{j\in O} x_i^j \le 1, \quad \forall i\in T, \\
& x_i^j \in \{0,1\}, \quad os_i^j \in \mathbb{R}, \\
& x_i^j \cdot (os_i^j - vs_i^j) \ge 0, \quad \forall i\in T, j\in O, \\
& x_i^j \cdot (ve_i^j - d_i - os_i^j) \ge 0, \quad \forall i\in T, j\in O, \\
& os_m^j + d_m + \mathrm{tran}\left(\Delta g(OW_m^j, OW_n^j)\right) \le os_n^j, \\
& \quad \forall j\in O, \forall m,n\in T \text{ with } x_m^j = x_n^j = 1
\end{aligned}$$

Methodology

Framework Overview

The overall HCPS architecture integrates heuristic candidate-space shaping with policy-based selection. The workflow comprises five key modules that operate iteratively to construct the schedule.

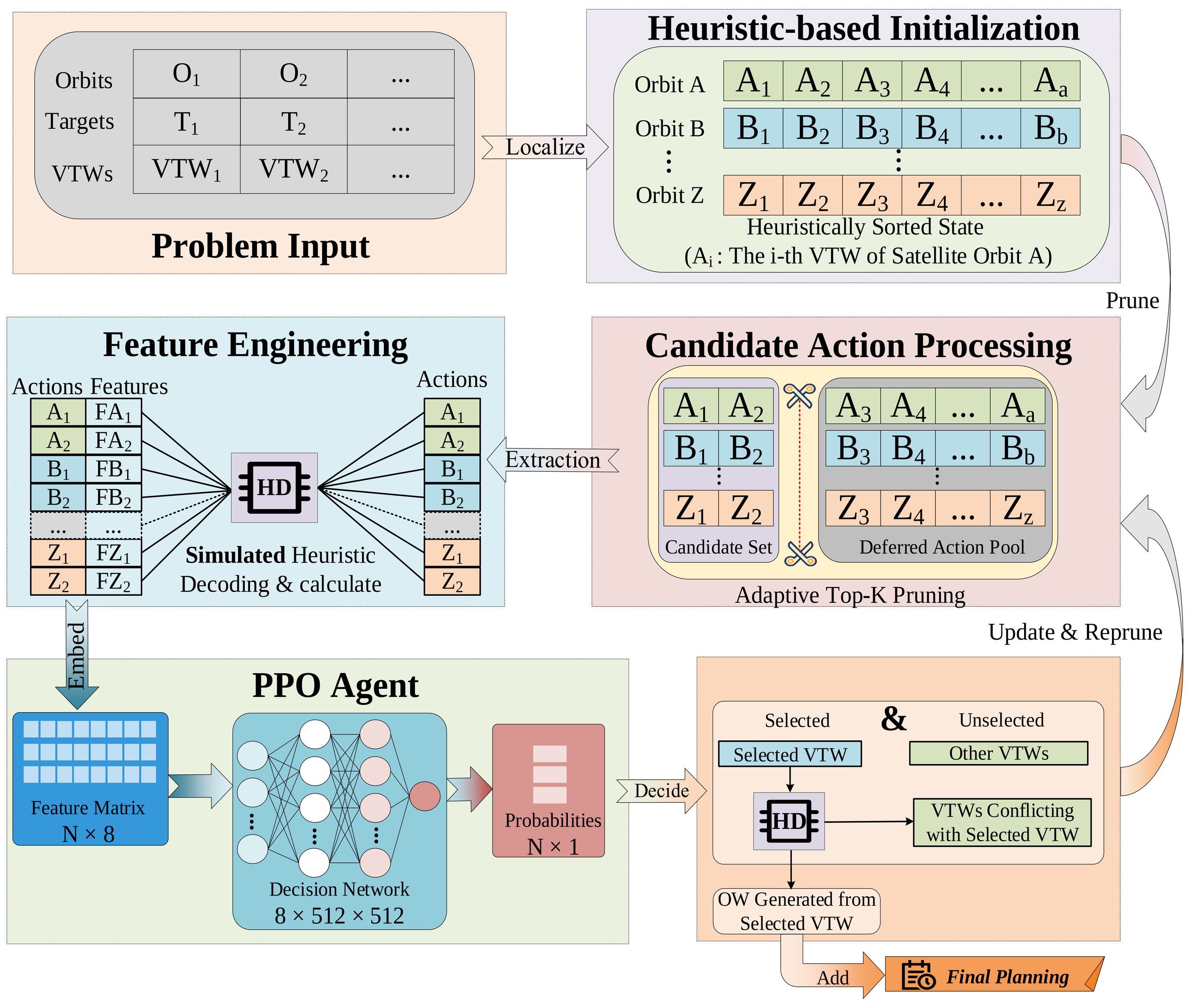

Figure 2: Overall framework of HCPS.

HBI — Heuristic-Based Initialization

(1). Conflict-driven orbit splitting

Each orbit O_j is recursively split at any gap Δt_k between consecutive VTWs satisfying $\Delta t_k > \tau_{max}^j$ yielding independent sub-orbits with no cross-conflict. The set of sub-orbits represents the smallest orbital units within the algorithm and is designated as virtual sub-orbits for independent processing.

(2). Heuristic intra-orbit ordering

Assign priority score to each VTW: $G_i^j = \alpha\left(1 - \frac{ve_i^j - st_j}{et_j - st_j}\right) + \beta\left(1 - \frac{vs_i^j - st_j}{et_j - st_j}\right)$

with $\alpha = 1$ (EDF, deadline pressure) and $\beta = 0.2$ (start-time preference). Sort descending by G_i^j to form a temporally coherent initial state.

CAP — Candidate Action Processing

Resource ratio for orbit O_j at step t:

$$r_j(t) = \frac{T_{\text{need}}^j(t)}{T_{\text{avail}}^j(t)}, \quad T_{\text{need}}^j = N_{VTW}^j \cdot \bar{t}_{\text{occ}}, \quad T_{\text{avail}}^j = et_j - st_j - t_{\text{used}}$$

Adaptive K per orbit:

$$K_j(t) = \max\left(2, \lfloor r_j(t) \rfloor\right)$$

Candidate set (Top-K per orbit based on priority scores G_i^j:

$$C_t = \bigcup_j \mathrm{TopK}_{a\in A_t^j} G_i^j, \qquad |C_t| = \sum_j K_j(t) \ll |A_t|$$

Deep Feature Extraction

For each retained candidate, invoke the Heuristic Decoder (HD) in simulation mode and extract six raw features:

- Transition time overhead (normalized by $Tran_{max}$)
- Conflict count
- Target kill count
- Resource pressure (remaining time / remaining VTWs)
- Candidate rank (position in Top-K list, normalized)
- Remaining opportunities (per target, normalized)

These raw features are then normalized within C_t and passed to the PPO agent.

HD — Heuristic Decoder

(1). Simulation mode

Copies the schedule, tentatively inserts the candidate OW, and returns the resulting feature vector without altering the live schedule.

(2). Execution mode

Inserts the OW into the live schedule, computes the reward, and removes the scheduled target and conflicting VTWs.

PPO Agent — Learning on Candidate Space

(1). State & Action Representation

The state is implicitly encoded in the feature vectors of candidates in C_t.

Action space: C_t, each action corresponds to selecting a specific VTW.

(2). Input Features (8-dimensional per candidate)

- Six features from CAP
- Two additional features: target profit p_i and required duration d_i.

(3). Network Architecture

- Actor: Shared feature extraction layer + softmax over K candidates:
- Critic: Independent per-candidate transformation → average pooling → scalar state value.

(4). Reward Function (Hierarchical Design)

To provide dense learning signals while preserving focus on the primary objective, we design a hierarchical reward function that balances four components:

$$R_{all} = R_{comp} + R_{kill} + R_{time} + R_{chance}$$

where

$$R_{comp} = \omega_1 p_i, R_{kill} = -\omega_2 \sum_i p_i k_i$$

$$R_{time} = -\omega_3 \cdot \frac{tran - Tran_{\min}}{Tran_{\max} - Tran_{\min}},$$

$$R_{chance} = \omega_4\left(1 - \frac{chance}{Chance_{\max}}\right)$$

Results

Effect of Adaptive Top-K Pruning

We evaluate fixed K values $\{1, 3, 5, 10\}$ and the proposed adaptive K strategy (K_{ada}) on three representative scales (700, 1000, 1500 targets). Figure 3 illustrates the training convergence and critic loss patterns. Smaller K values lead to faster initial convergence due to the reduced action space, but often saturate at lower quality. Larger K values enable more sustained improvement yet introduce higher variance and slower training. The adaptive K strategy achieves the best balance: it retains sufficient candidates when resources are tight while keeping the action space compact when possible, yielding superior solution quality with competitive efficiency. Based on these results, we adopt the adaptive K strategy for all subsequent experiments.

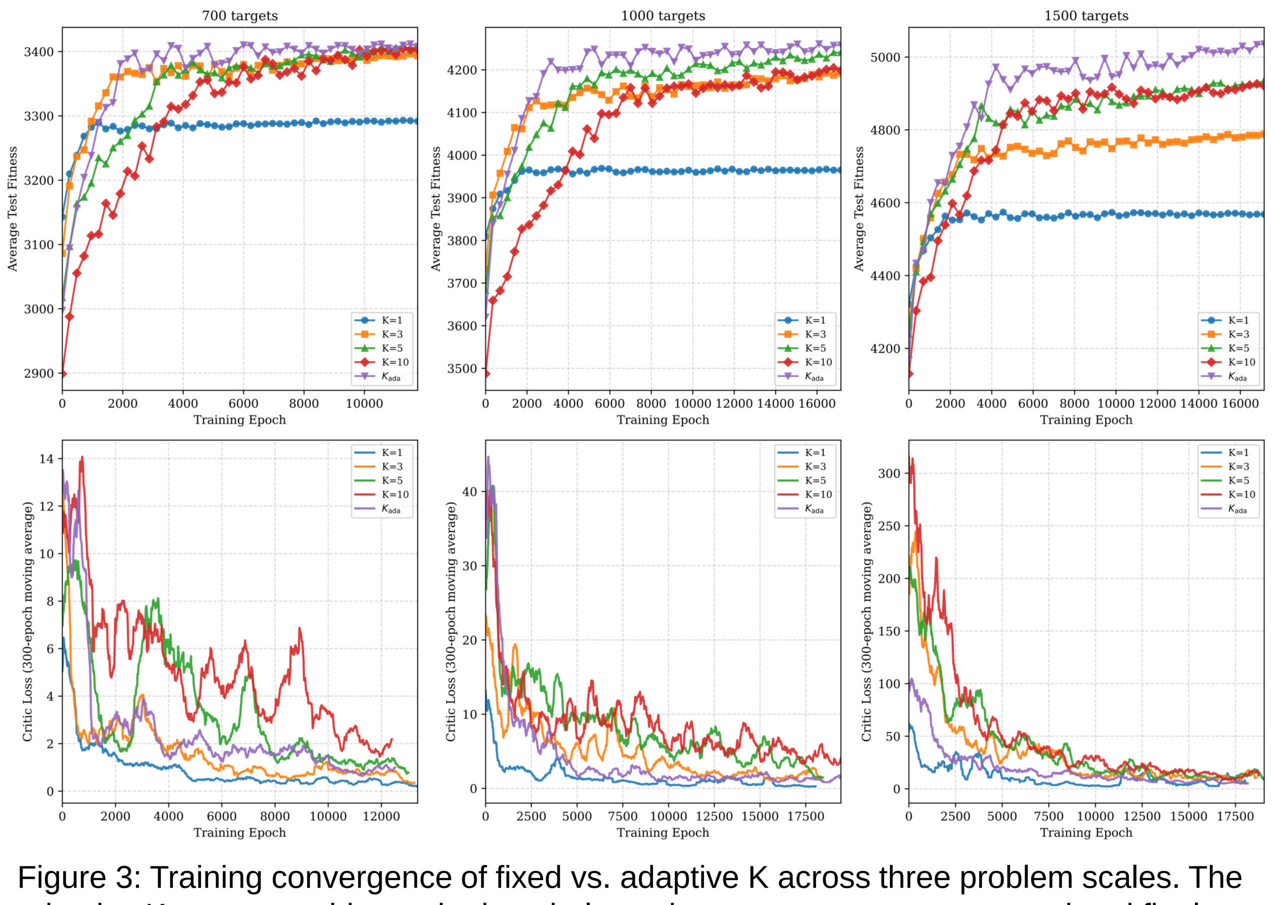

Figure 3: Training convergence of fixed vs. adaptive K across three problem scales. The adaptive K strategy achieves the best balance between convergence speed and final solution quality.

Comparison with State-of-the-Art Algorithms

We compare HCPS with four state-of-the-art baselines: GRILS, A-ALNS, GA, and RL-GA. All algorithms are evaluated on the same dataset and hardware, with parameters tuned to their optimal settings as reported in the original papers.

Figure 4 presents the target completion rate (a) and runtime (b, log scale) across all 15 problem scales (100–1500 targets).HCPS achieves 100% completion for instances up to 400 targets, and consistently yields the highest completion rates for scales 500–1500. In terms of efficiency, HCPS reduces runtime by an average of 84.8% compared to GA and 91.0% compared to GRILS for scales exceeding 300. Statistical significance is verified by Wilcoxon test ($p < 0.05$) for scales ≥ 500.

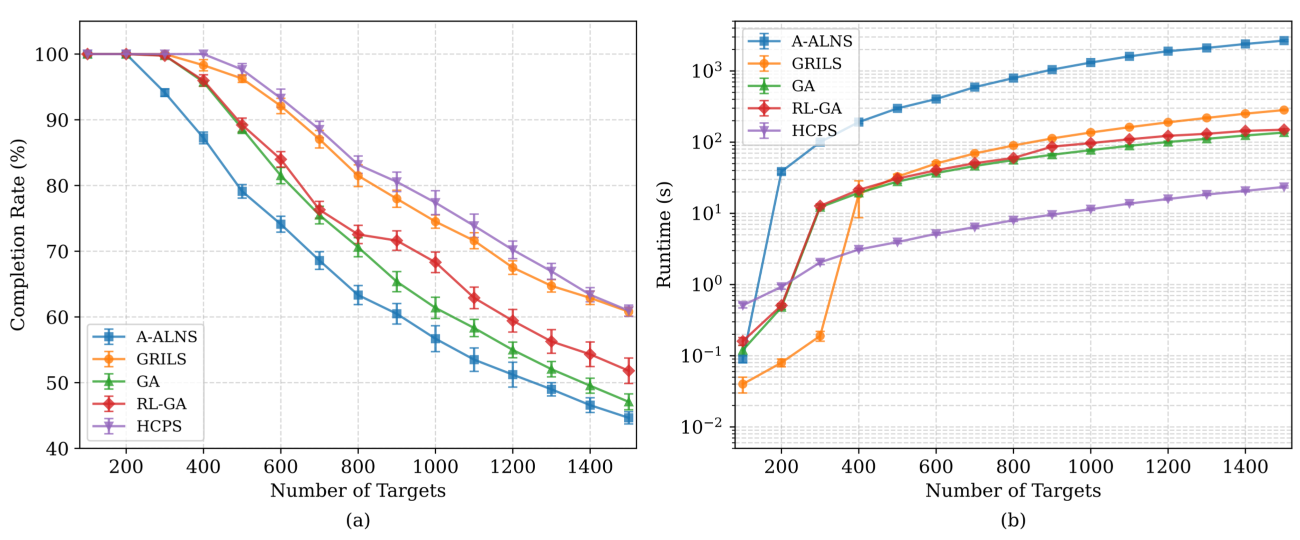

Figure 4: Comparison of (a) target completion rate and (b) computational time across 15 problem scales.

Ablation Studies

We conduct ablation experiments to quantify the contribution of each component in HCPS, including all engineered features (from CAP) and the hierarchical reward components. The full model (HCPS-Full) includes all features and the complete reward. We systematically remove each component and evaluate the impact on solution quality across three problem scales (700, 1000, 1500 targets). Table 1 reports the relative completion rate F_s normalized to the full model (set to 100%) when individual components are omitted.

The results indicate that the primary objective **target completion** has the largest impact, with its removal reducing performance by over 13%. Among the reward components, **kill penalty** also plays a critical role, causing a 6–8% drop when omitted. Among the features, **Target kill count** has the strongest influence. The relative importance of these components remains broadly consistent across problem scales, though the impact of removing components tends to be slightly smaller on larger scales.

Removed Component	700 targets	1000 targets	1500 targets
None (Full model)	100.00	100.00	100.00
Target completion (reward)	85.47	86.22	86.47
Kill penalty (reward)	92.15	93.44	93.72
Target kill count (feature)	94.60	94.24	95.44
Orbit resource pressure (feature)	97.07	96.85	97.40
Remaining opportunities (feature)	98.26	98.30	98.82
Urgency (reward)	98.46	98.71	99.15
Transition time overhead (feature)	98.65	98.83	99.61
Conflict count (feature)	99.07	99.85	99.79

Table 1 Ablation results: relative completion rate (F_s) when removing key components (normalized to full model = 100%).

Conclusion and Future Work

Conclusion

This paper proposed HCPS, a hybrid framework that treats domain heuristics as structural shapers of the search space for combinatorial optimization with expensive per-step evaluation.

The effectiveness of HCPS stems from two key factors. First, adaptive Top-K pruning reduces expensive decoding calls. The saved budget enables finer-grained analysis on retained candidates, extracting richer features that capture downstream consequences. These features provide the PPO agent with informed signals for value estimation. In essence, HCPS shifts cost from breadth to depth.

Second, the pruning preserves temporal causality. The priority score G_i^j encodes this via EDF and start-time preference, and Top-K operates within this sorted order. Thus, C_t maintains temporal coherence—VTWs close to the current schedule are kept, while disjoint options are pruned. Preserving temporal structure keeps the action space physically meaningful, enabling policies that respect the natural observation progression.

Future Work

Future work will incorporate uncertainty-aware features to improve robustness in noisy real-world environments, extend the temporal causality mechanism to cross-interval rescheduling for dynamic re-planning across multiple horizons, and transfer the HCPS paradigm to other large-scale engineering scheduling problems with expensive per-step evaluation to demonstrate its generalizability beyond agile Earth observation satellite scheduling.

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Availability

Code & data available upon request: gaopf@bupt.edu.cn

Paper Link Supplementary Materials